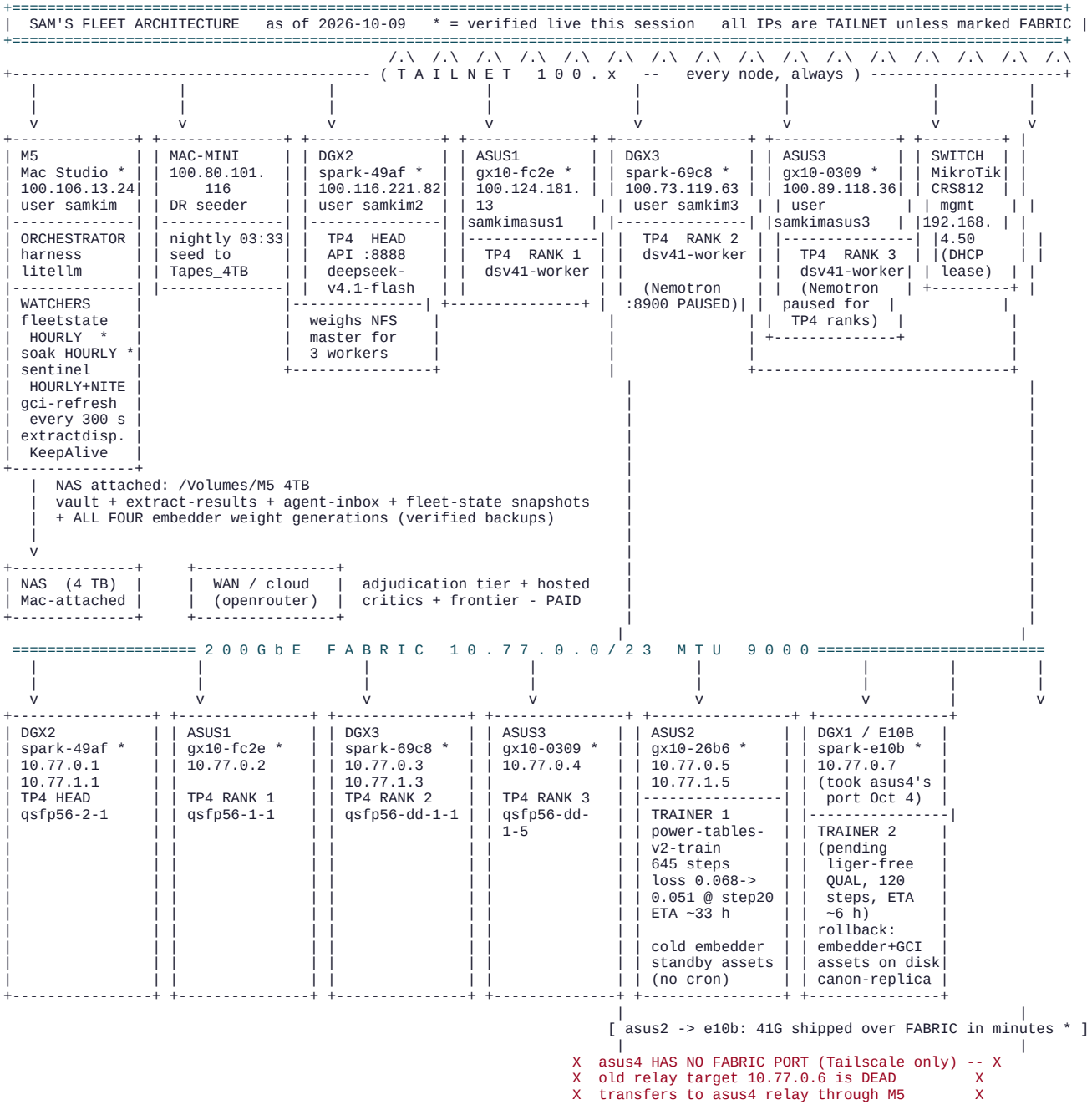
FLEET ARCHITECTURE - topology, tailnet + fabric planes

Sam's fleet | generated 2026-10-09 by m5-opencode from live-verified state | IPs: Tailscale 100.x unless marked | * = exercised this session



SSH ACCESS MATRIX + SERVICE FLOWS

Sam's fleet | generated 2026-10-09 by m5-opencode from live-verified state | IPs: Tailscale 100.x unless marked | * = exercised this session

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+=============================================================================================================================+
|  SSH ACCESS MATRIX - from whom, to what, as whom, over which path            * = exercised and verified 2026-10-09   |
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  FROM (who)               TO (what)                        AS (user@)                    PATH / METHOD            STATUS
  ----------------------   ------------------------------   ---------------------------   ----------------------   --------
  M5  (Mac Studio)         ALL SEVEN fleet nodes:                                                                          *
    orchestrator + all      -> DGX2  (TP4 head)               samkim2@100.116.221.82       tailnet, pubkey           *
    watchers + dispatch     -> ASUS1 (TP4 r1)                 samkimasus1@100.124.181.13   tailnet, pubkey           *
                            -> DGX3  (TP4 r2)                 samkim3@100.73.119.63        tailnet, pubkey           *
                            -> ASUS3 (TP4 r3)                 samkimasus3@100.89.118.36    tailnet, pubkey           *
                            -> ASUS2 (trainer 1)              samkimasus2@100.68.133.1     tailnet, pubkey           *
                            -> ASUS4 (embedder/GCI/canon)     samkimasus4@100.100.116.82   tailnet, pubkey           *
                            -> DGX1  (trainer 2)              samkim@100.81.201.24         tailnet, pubkey           *
    M5 is the SSH hub: every probe, dispatch, deploy and transfer originates here or relays through it.

  ASUS2 (trainer 1)        -> DGX1  (trainer 2)              samkim@10.77.0.7             FABRIC, pubkey            *
                              (used to ship 41G: llamafactory image + 8B base + dataset)

  ASUS4 (serving)          (none outbound -- canon primary, embedder, GCI all serve; box initiates nothing)            -

  DGX1/E10B                -> ASUS4 attempted via ssh -A agent-forwarding                  FAILED: Mac agent key     X
                              not forwarded. All e10b->asus4 bulk moves relay through M5 tar-pipe.

  ASUS2                    -> ASUS4 via fabric relay 10.77.0.6                             DEAD since Oct 4:         X
                              (asus4 gave its switch port to dgx1)                          asus4 is Tailscale-only

  MAC-MINI                 -> Tapes_4TB (USB)                local                       nightly rsync 03:33       -

  EVERYONE                 -> NAS /Volumes/M5_4TB            only via M5                 box is Mac-attached       -


+=============================================================================================================================+
|  SERVICE FLOWS - who calls what, how often                                                                                  |
+=============================================================================================================================+

  FLOW  FROM        TO                       CADENCE      WHAT / WHY
  ----  ----------  -----------------------  -----------  ------------------------------------------------------------------
  F1    M5          DGX2  :8888              hourly       /v1/models - engine liveness (probe key tp4_dsv41)
  F2    M5          ASUS4 :8900              hourly       /v1/models - canon judge liveness (asus4_canon) +
                                                              soak alerts canon_down when dark
  F3    M5          DGX1  :8901              hourly       /v1/models - canon replica liveness (dgx1_canon_replica)
  F4    M5          ASUS4 :8800/healthz      hourly       embedder identity + readiness (model, device, routes)
  F5    M5          ASUS4 :8810/health       hourly       GCI liveness (401 = alive, bearer-guarded, no token sent)
  F6    M5          all 7 nodes              hourly       ssh 'docker ps' - container truth; deltas -> CHANGES.md
                                                              + change-mail to cursor-cr, m5-opencode, m4opencode
  F7    M5          ASUS4 :8810              every 300 s  launchd gci-refresh: index upkeep (launchd key
                                                              com.samkim.harness-gci-refresh)
  F8    M5          ASUS4 :8900              hourly+      sentinel: claims re-verify + nightly gold drift
                                                              (CANON hard-coded in 4 code paths)
  F9    M5          ASUS2 + ASUS4            continuous   extractdispatch (KeepAlive): PDF extract workers
                                                              over ssh, ~/extract-jobs/{incoming,out,err}
  F10   any client  DGX2  :8888              on demand    THE ENGINE: deepseek-v4.1-flash, vision + 1M ctx
  F11   orchestr.   ASUS4 :8800              on demand    spark_embed lane (config keys spark_embed/spark_embedder)
  F12   GCI (asus4) ASUS4 :8800  (localhost) internal     encoder pinned to 127.0.0.1:8800 by policy
                                                              (harness/gci/encoder.py, allowlist available)
  F13   MAC-MINI    Tapes_4TB                nightly 03:33 DR: extract-results + agent-inbox + canon-gate
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PER-NODE DETAIL - state, windows, rollback


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+=============================================================================================================================+
|  PER-NODE SERVICE DETAIL + STATE / WINDOWS / ROLLBACK                                                                       |
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  TP4 CLUSTER (4 boxes, committed) -- SGLang TP4, deepseek-v4.1-flash, 552B MoE, vision + 1M ctx + tool calls
    DGX2   head, API :8888, weights on local NVMe, exports /srv/models via nfsd to ranks
    ASUS1/DGX3/ASUS3   dsv41-worker ranks; read weights over nfs4 (ro)
    ASUS3 note: Nemotron :8900 (independent critic) PAUSED while its GPU serves TP4 -- reviewer lane is
                dark-by-choice; registry reports "healthy" (no liveness check in Worker.status())

  ASUS4 -- THE SERVING BOX (Tailscale-only, no fabric port; canon + embeddings + GCI live here)
    :8900  canon-qwen38-27b      sglang, qwen3.8-27b-fp8 dense, THE judge (REJECT authority), determinism 5/5
    :8800  bge-m3-embed.service  FAE v4 fine-tune behind legacy label bge-m3-cr-tapes-v1, GET /healthz
                                 RELOCATED from e10b 2026-10-09; cutover gate: 8/8 vectors BIT-IDENTICAL
    :8810  harness-gci.service   Global Code Intelligence; 401=alive; same token as before (clients unchanged)
           extract_worker        PDF text extraction for dispatch lane
    mem: canon ~66G + pair ~9G, ~33G headroom

  ASUS2 -- TRAINER 1 (proven liger-free box; the 1,020-step v1 run lived here)
    RUNNING: power-tables-v2-train -- THE FIRST REAL v2 STUDENT EVER (all prior "v2" artifacts were
             mislabeled v1 merges) -- 645 steps, ~186 s/step, ETA ~33 h, loss falling, ETA gate next
    COLD:   ~/embedder-rehearsal -- verified weight copy + venv + service (no cron, starts only by hand)
    disk 68% (283G free) after 535G verified-duplicate reclaim

  DGX1/E10B -- TRAINER 2 (conditional)
    RUNNING: e10b-ligerfree-qual -- 120 steps, use_liger_kernel FALSE, artifacts persistent, ETA ~6 h
             the Oct 6-7 freezes all ran liger ON and died by ~step 42; 120 clean steps = 3x past it
    WINDOW OPEN: canon-replica STOPPED for the qual -> canon is SINGLE-REPLICA on asus4 until it ends
    ROLLBACK ON DISK (stopped+disabled): bge-m3-embed + harness-gci units, /data/categoryrank weights,
             /data/harness-gci index, ~/harness-gci app -- re-enable + repoint 4 configs to undo
    CRASH SUSPECTS (both recorded): 1) liger -- documented, all 3 freezes ran it
                                    2) HDMI hotplug -- Sam's report, crashes on unplug TODAY;
                                       dummy plug + GUI-off queued after verdict; DO NOT unplug mid-run

  M5 -- CONTROL PLANE: orchestrator, litellm, all launchd watchers, NAS, the ONLY SSH hub
  MAC-MINI -- DR: nightly 03:33 seed of extract-results / agent-inbox / canon-gate to Tapes_4TB
  NAS -- vault (source of record), extract-results (pipeline.db, catalog.db, soak.csv),
         agent-inbox (fleet mail), fleet-state snapshots, ALL FOUR embedder weight generations

  SINGLE POINTS OF FAILURE (accepted, by design of this architecture)
    canon judge           -> asus4 alone during training windows (replica returns between windows)
    embedder + GCI        -> asus4 alone; cold standby assets on asus2; NAS backups of every generation
    M5 itself             -> all watchers, dispatch, and the only SSH hub; factory keeps filling vault if down

  THE FLYWHEEL (why two trainers)
    burns land -> catalog.db claims (87K quote-native) -> build_lora_pairs_v2 -> train on 2 boxes
    -> gate_v2 (f1>=0.90, precision>=0.95) -> promote -> serve -> more validated pairs -> repeat
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